(Please write your Enrollment Number) Enrollment No. 04801172022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech, Semester:7th
(December, 2025)

Subject Code: BAI 403	Subject: Computer Vision
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
	a) Explain the concept of a Loss Function in the context of Linear Classification. Give one example of a commonly used loss function.	
	b) What is Parameter Sharing in Convolutional Neural Networks (CNNs)? Why is it crucial for efficiency?	
	c) Differentiate between the three components of the Transformer architecture: Key (K), Value (V), and Query (Q).	
	d) Briefly describe the difference between Semantic Segmentation and Object Detection.	
	e) What is an Adversarial Example in Machine Learning? How is it used to evaluate the robustness of a classifier?	
	f) Explain the primary purpose of Batch Normalization. Where is it typically placed within a CNN architecture?	
	g) What is the fundamental distinction between Soft Attention and Hard Attention mechanisms in Vision & Language models?	
	~~h) What is an Adversarial Example in Machine Learning? How is it used to evaluate the robustness of a classifier?~~	
	UNIT I	
Q2	a) Explain the Data-Driven Approach to Computer Vision in contrast to earlier, hand-engineered approaches. Why has the data-driven approach dominated the field? (5 Marks) b) Describe the working principle of the K-Nearest Neighbors (KNN) classification algorithm. Discuss two major practical challenges or limitations faced by KNN when used for large-scale image classification tasks. (5 Marks)	(10)
Q3	a) Explain the concept of an Activation Function in a Neural Network. Compare and contrast the Sigmoid and ReLU (Rectified Linear Unit) activation functions, highlighting why ReLU is often preferred in hidden layers of deep networks. (5 Marks) b) Define the problem of Overfitting in neural networks. Describe the technique of Early Stopping and explain how it serves as a regularization method to mitigate overfitting. (5 Marks)	(10)

	UNIT II	
Q4	a) Compare and contrast the architectures of LeNet and VGG-16. Discuss the architectural design philosophy of VGG-16 (e.g., use of small kernels) and explain why it is computationally expensive compared to modern networks. (6 Marks) b) What is Transfer Learning? Explain the two primary ways Transfer Learning is implemented (e.g., using a CNN as a fixed feature extractor vs. fine-tuning) for a new task. (4 Marks)	(10)
Q5	a) Describe the function of the following layers in a modern CNN, using suitable examples: i) Convolutional Layer ii) Pooling Layer (Max-Pooling) iii) Fully Connected Layer (6 Marks) b) Define the concepts of L1 and L2 Regularization. How do these techniques help in preventing overfitting during the training of a CNN? (4 Marks)	(10)
	UNIT III	
Q6	a) Describe the typical Encoder-Decoder architecture used for Image Captioning. Explain the role of the CNN (as the Encoder) and the Recurrent Neural Network (RNN) or Language Model (as the Decoder). (5 Marks) b) Explain the operation of the Attention Mechanism in the context of Image Captioning. How does Attention allow the decoder to focus on specific regions of the input image while generating the corresponding word? (5 Marks)	(10)
Q7	a) Explain the process of Self-Attention within a single layer of a Transformer Encoder. Why is Self-Attention considered a breakthrough mechanism for modeling long-range dependencies in sequence data? (5 Marks) b) Describe how the Transformer architecture, originally designed for sequence-to-sequence tasks, has been adapted for use in pure Image Recognition. Briefly mention one popular Transformer-based model used for image classification. (5 Marks)	(10)
	UNIT IV	
Q8	a) Explain the fundamental architecture and training objective of a Generative Adversarial Network (GAN). Clearly define the roles of the Generator and the Discriminator, and what the term "adversarial" signifies. (6 Marks) b) Discuss the critical issue of Biases in Image Datasets. Provide an example of how dataset bias can lead to unfair or incorrect decision-making in Computer Vision applications like Face Recognition. (4 Marks)	(10)
Q9	Define Semantic Segmentation and justify why traditional Convolutional Neural Networks are unsuitable for this task.	(10)